

Decode the Chart

Name: Pranayaa S Roll No: 30 Reg No: 192524110

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

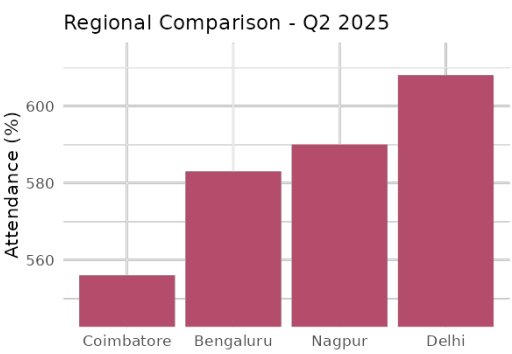


Chart 1

Expense Breakdown by Category (13 categories)



Chart 2

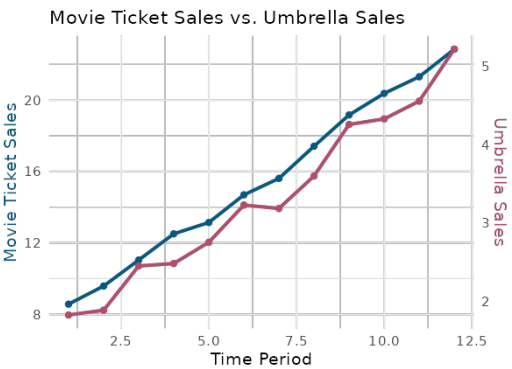


Chart 3

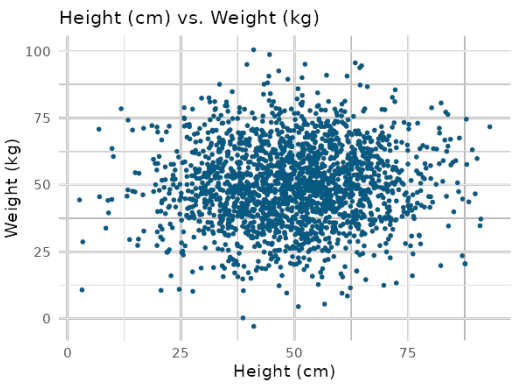


Chart 4

Decode the Chart — Set 30 of 40

What is each chart trying to tell me, and do I trust it?

Chart 1: Attendance by city (bar chart)

Delhi looks miles ahead and Coimbatore looks far behind, with Delhi's bar about three times taller. But the y-axis starts around 545, not zero. The actual values are roughly 557 for Coimbatore, 583 for Bengaluru, 590 for Nagpur and 608 for Delhi, so the real gap is only about 9%. The chopped axis is doing all the work. On top of that, the label says "Attendance (%)" while every value is over 500, which can't be right for a percentage. I don't trust this one. Starting the axis at zero and fixing the label would show the honest picture.

Chart 2: Expense breakdown (pie chart)

This show how expenses are split, but with 13 slices, no labels and no legend, I can't tell what any slice stands for. Several are thin slivers and many of the greens and blues blend together, so even ranking the top three is guesswork. I don't trust it for comparisons. A sorted bar chart with values would be much easier to read, or the top five or six categories could be kept with the rest grouped as "Others".

Chart 3: Movie tickets vs umbrella sales (dual-axis line chart)

The two lines move almost perfectly together, so it looks as if movie tickets and umbrellas are connected. They aren't, at least not in any way that makes sense. Each line has its own y-axis (about 8 to 22 on the left, 2 to 5 on the right), and the scales have been chosen so the lines overlap. Both are simply rising over time, perhaps because of population growth or inflation. This is a spurious correlation: the match comes from the axis choice, not from a real link. Separate panels or a shared scale such as % change would be more honest.

Chart 4: Height vs weight (scatter plot)

This should show whether taller people weigh more, but there are so many solid dots (1,500+) that the middle is one dark blob, and I can't see where the data really piles up. The numbers also look off: most heights fall between 25 and 75 cm and some weights are close to 0 kg, which isn't realistic for people. There's no upward trend either, even though height and weight normally go together. Transparent dots or a hexbin plot would fix the overplotting, but the data itself needs checking first.